

Cell Datasheet - Long-Endurance Drone Battery

VBF-T8R1NOFORCE-DSH-01 | case t8_r1_noforce | 2026-08-25 | all values from bda tool output; not a physical cell

General

Field	Value	Source
Cell designation	VBF-T8R1NOFORCE-CELL-R6V9	delivery index
Electrochemical system	NMC811 / graphite (pouch geometry)	Chen2020 parameter set
Rated capacity	5.2725 Ah (1C discharge, 25 C, DFN)	r6_v9_1c_dfn.json
Nominal voltage	3.933 V (1C midpoint)	r6_v9_energy.json
Voltage window	2.5 - 4.2 V	parameter set
Rated energy	19.4147 Wh	r6_v9_energy.json
Energy density (gravimetric)	497.968 Wh/kg (contract; electrolyte excluded)	r6_v9_energy.json
Energy density (volumetric)	854.72 Wh/L	r6_v9_energy.json
DC resistance (formulation)	0.7275 mohm	r6_v9_energy.json dcr_ohm
Power density (formulation)	150.5 kW/kg	r6_v9_energy.json

Physical

Field	Value	Source
Electrode dimensions	65 mm height x 1891 mm width; area 0.122915 m2	parameter set / r6_v9_energy.json
Stack thickness	184.8 um (pos 75.6 + sep 10 + neg 85.2 + Al 8 + Cu 6)	r6_v9_energy.json
Cell mass (contract)	38.99 g (electrolyte excluded per calc-energy)	r6_v9_energy.json mass_kg
Cell mass incl. electrolyte	48.54 g (electrolyte 9.55 g, literature density)	BOM sheet

Electrical performance (simulated, DFN)

Condition	Result	Source
1C discharge, 25 C	5.2725 Ah; T_max 299.18 K	r6_v9_1c_dfn.json
5C discharge, 25 C	5.2178 Ah; T_max 313.64 K; retention vs 1C = 0.9896	r6_v9_5c_dfn.json / r6_v9_retention.json
Continuous discharge rating	5C (requirement retention >= 0.90 - met at 0.9896)	r6_v9_retention.json
Fast charge capability	4C at 45 C: anode min +0.00635 V -> no plating; T_max 326.81 K	r6_v9_4c_charge45_plating.json

Environmental and safety

Field	Value	Source
Operating temperature (simulated)	25 - 45 C (1C/5C at 25 C; 4C charge at 45 C)	protocol runs, honest scope
Cooling requirement	h = 60 W/m2/K forced-air (ducted prop-wash)	r6_v9_kin_el.json
Lithium plating (4C charge)	none - anode min +0.00635 V (plated = false); thin margin +6.3 mV	r6_v9_4c_charge45_plating.json
Thermal runaway	not simulated (no abuse runs in this case)	honest N/A
Cycle life	not simulated	honest N/A (aging model present; not a task criterion)

Application note

Target: long-endurance drone. 5C discharge support (retention 0.9896) covers launch/climb bursts; energy density 497.97 Wh/kg (contract) with 38.99 g cell mass supports endurance-optimized packs. The design is negative-limited (N/P 0.641 capacity-density basis) - a deliberate trade for transport: the 85.2 μm graphite electrode keeps electrolyte-path polarization low at charge end, protecting the plating margin. Thin 6-8 μm current collectors and 10 μm separator carry the energy-density budget - aggressive engineering choices requiring pouch-fabrication capability.

Simulation honesty note: all values are DFN simulation outputs on the Chen2020 parameterization with design-parameter overrides; electrolyte transport parameters (κ 1.4 S/m, D 3.7e-10 m²/s) are literature values (Nyman 2008) substituted as formulation levers; electrolyte mass uses literature density 1.2 g/cm³. Physical-cell qualification (formation, mechanical integrity, cycle life) is out of scope.